AWS EC2 and EBS Volume Attachment Report

Step-by-step practical report for launching an EC2 Ubuntu instance, creating an Amazon EBS volume, attaching it to EC2, and verifying it through Linux terminal commands.

AWS Region	Asia Pacific (Mumbai) - ap-south-1
Instance Name	Ebs-connect
Instance ID	i-0b81e1e5bf5e41eb5
Operating System	Ubuntu 26.04 LTS
EBS Volume	vol-019e9869a679c8513, 5 GiB, gp3
Device / Mount Point	/dev/nvme1n1 mounted on /mnt/mydata

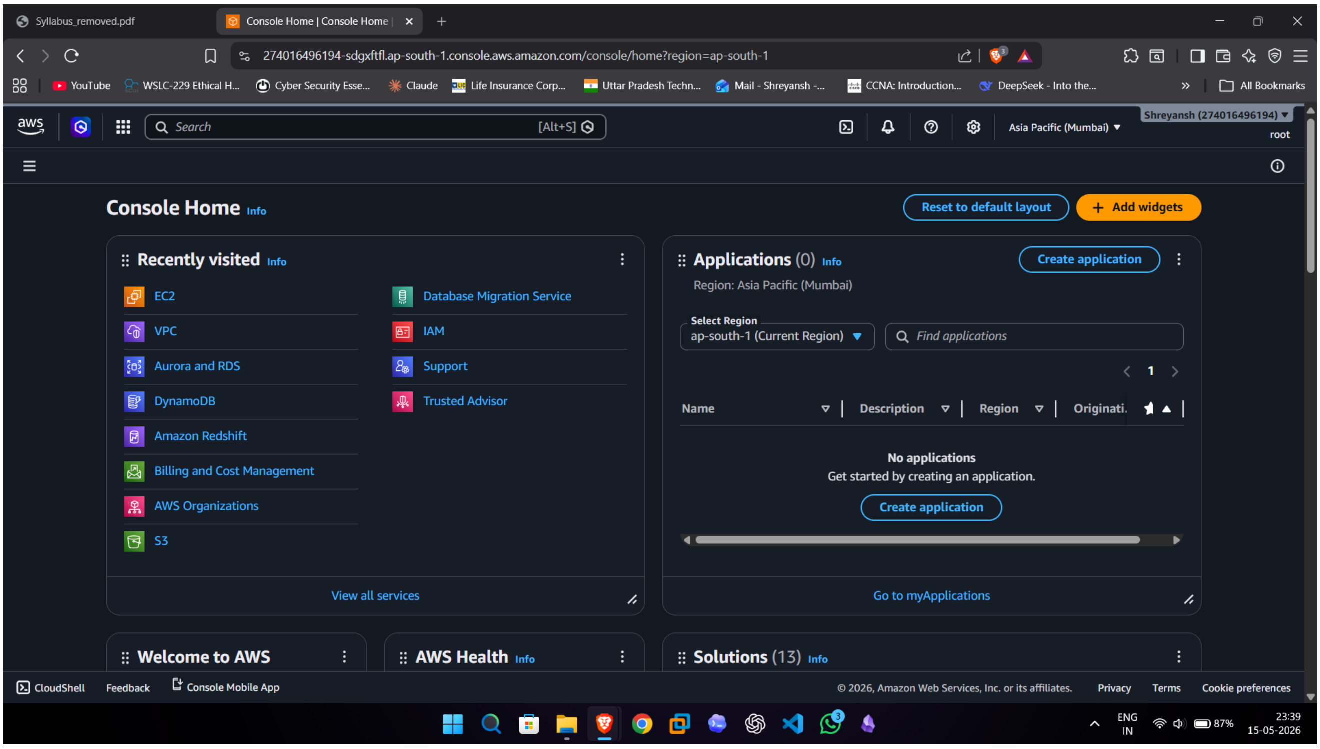
Practical Overview

- Launch an EC2 Ubuntu instance in AWS.
- Create a new key pair for SSH login.
- Configure security group rules for SSH, HTTP, and HTTPS.
- Create a 5 GiB gp3 Amazon EBS volume in the same availability zone.
- Attach the EBS volume to the running EC2 instance.
- Format, mount, and verify the EBS volume from Ubuntu Linux.

Implementation Steps

Step 1: Open AWS Console Home

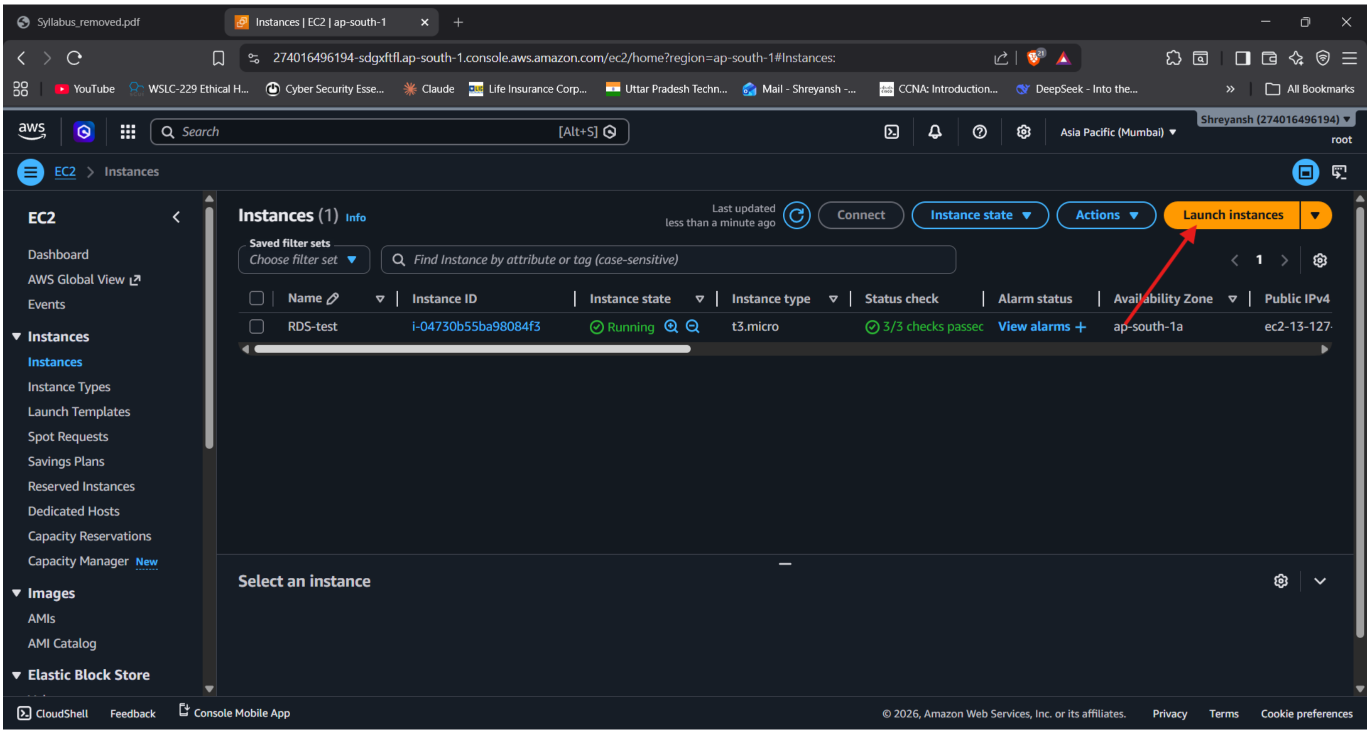
The AWS Console Home page was opened in the Asia Pacific (Mumbai) region.



Screenshot 2026-05-15 233941.png

Step 2: Open EC2 Instances Dashboard

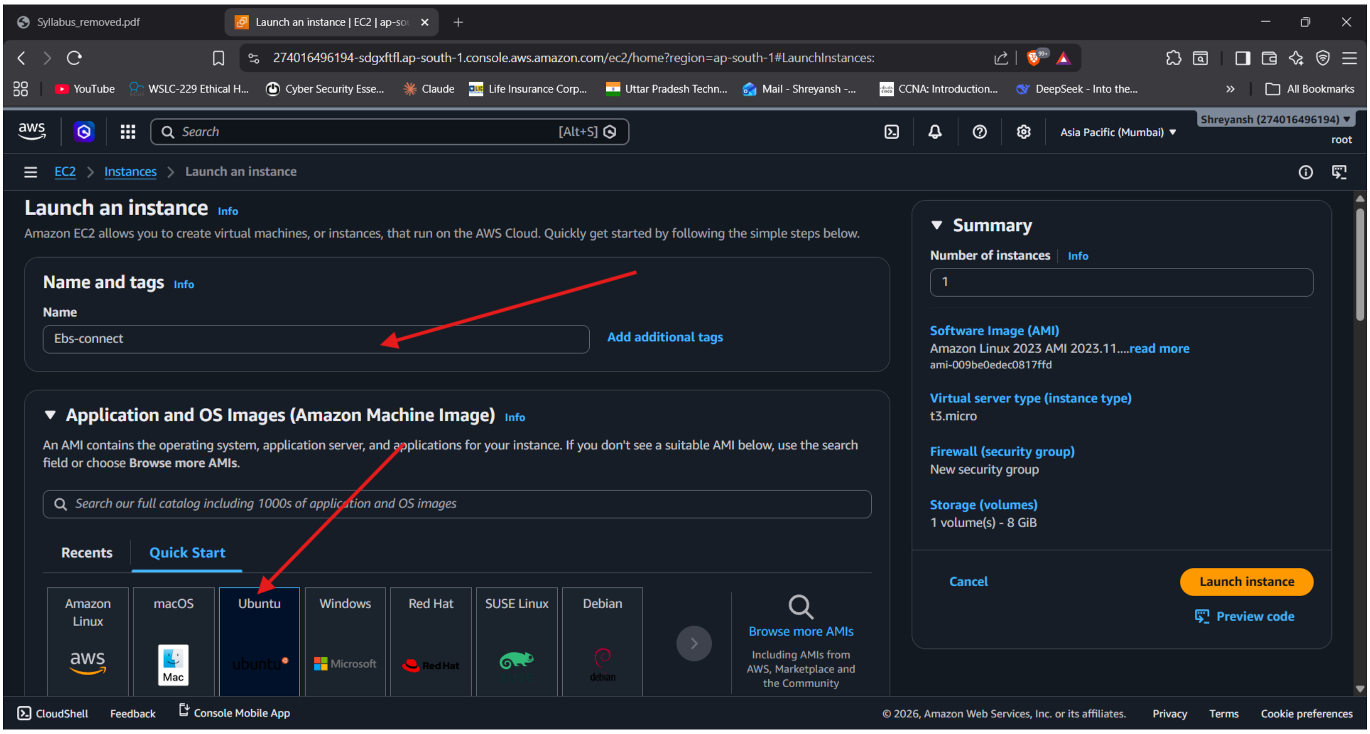
The EC2 Instances page shows the existing running instance and the Launch instances option.



Screenshot 2026-05-15 234022.png

Step 3: Configure Instance Name and AMI

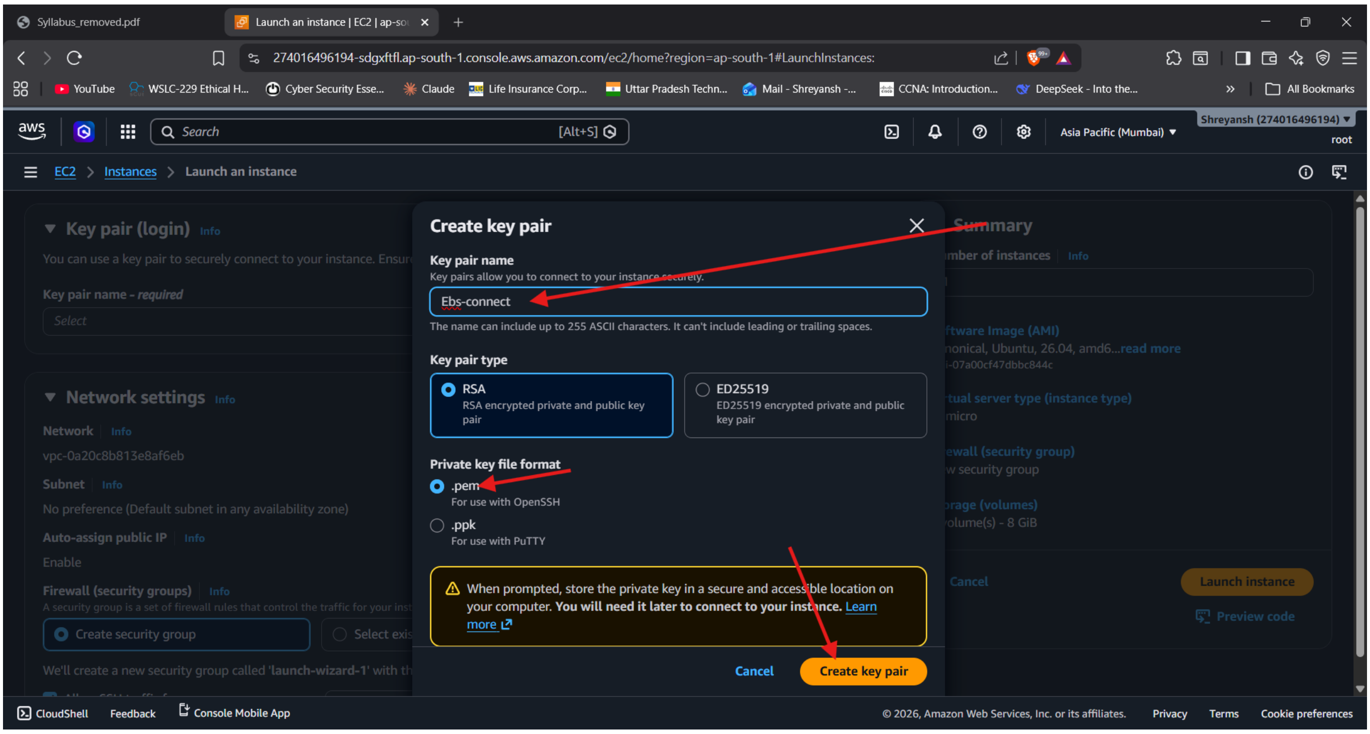
A new EC2 instance named Ebs-connect was created using an Ubuntu AMI and t3.micro instance type.



Screenshot 2026-05-15 234121.png

Step 4: Create Key Pair

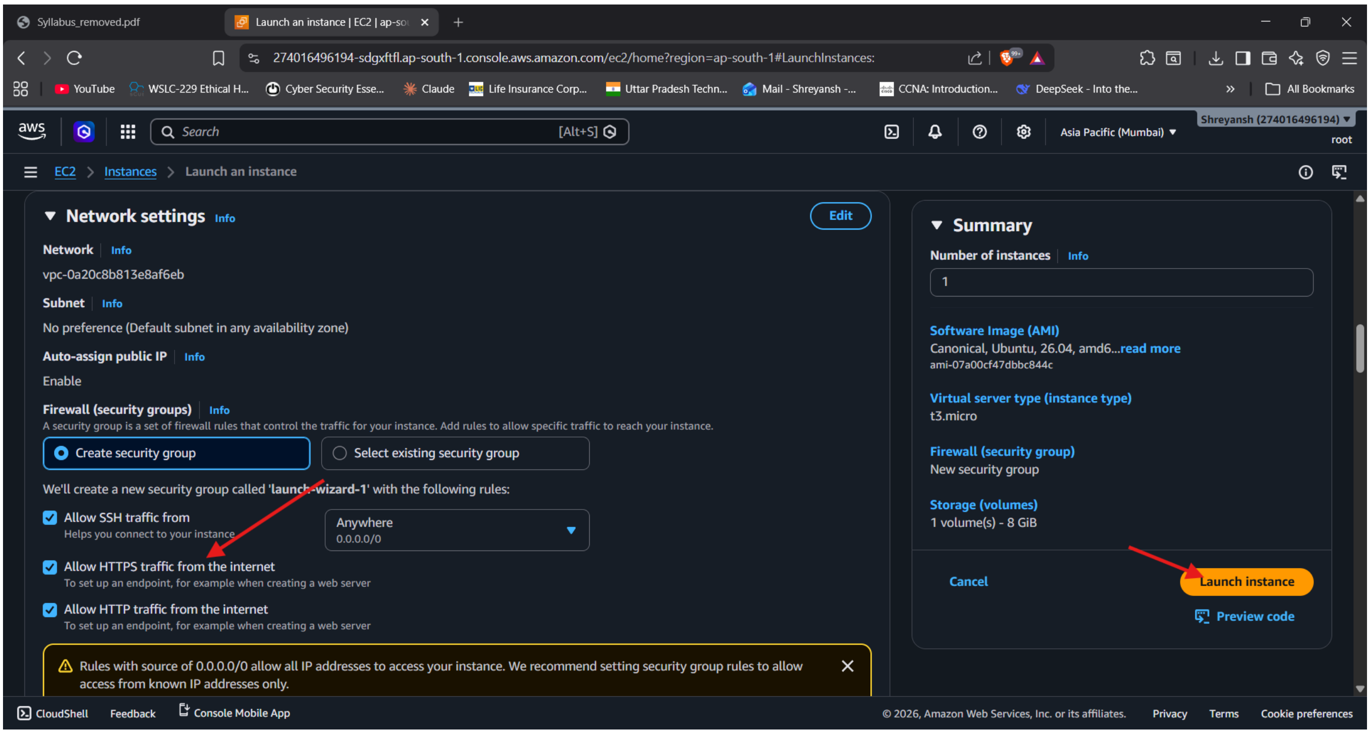
A new key pair named Ebs-connect was created using RSA and .pem format for SSH access.



Screenshot 2026-05-15 234307.png

Step 5: Configure Network and Launch Instance

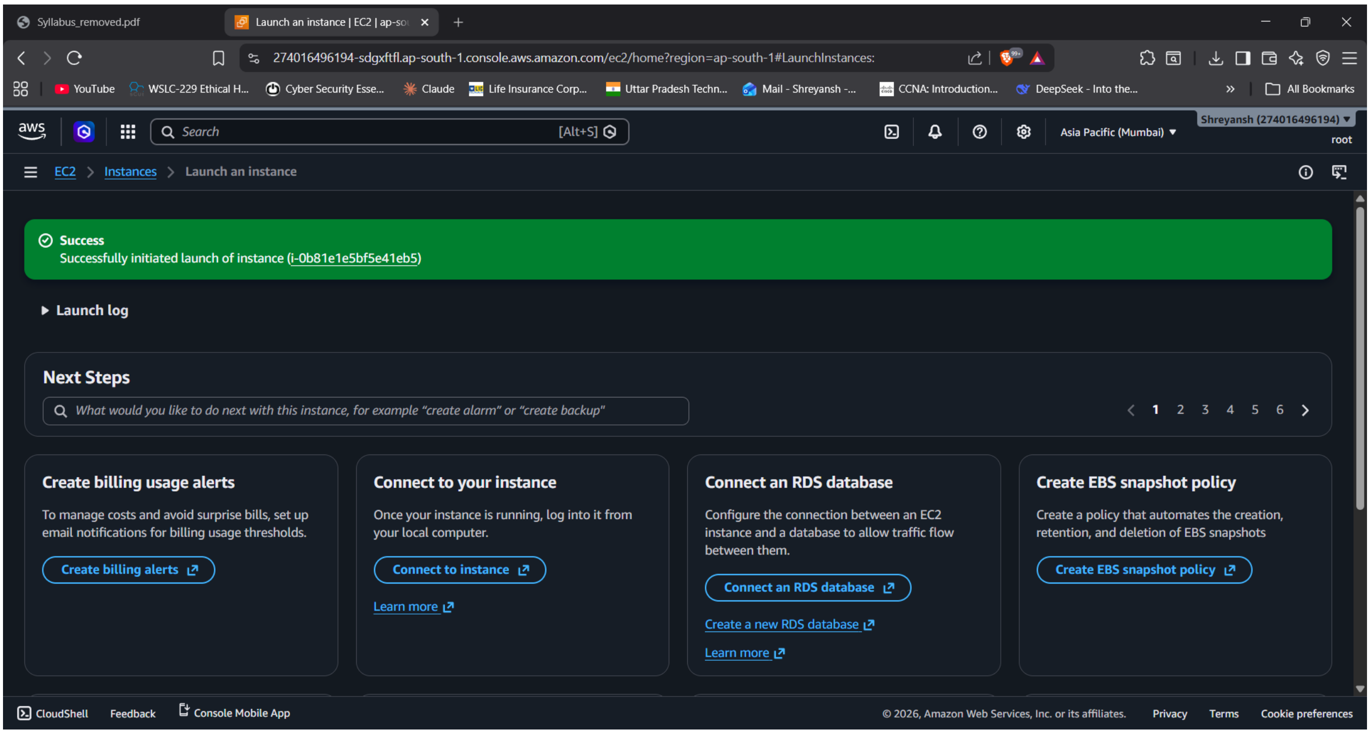
SSH, HTTP, and HTTPS rules were enabled, and the instance launch was started.



Screenshot 2026-05-15 234535.png

Step 6: Verify Successful Instance Launch

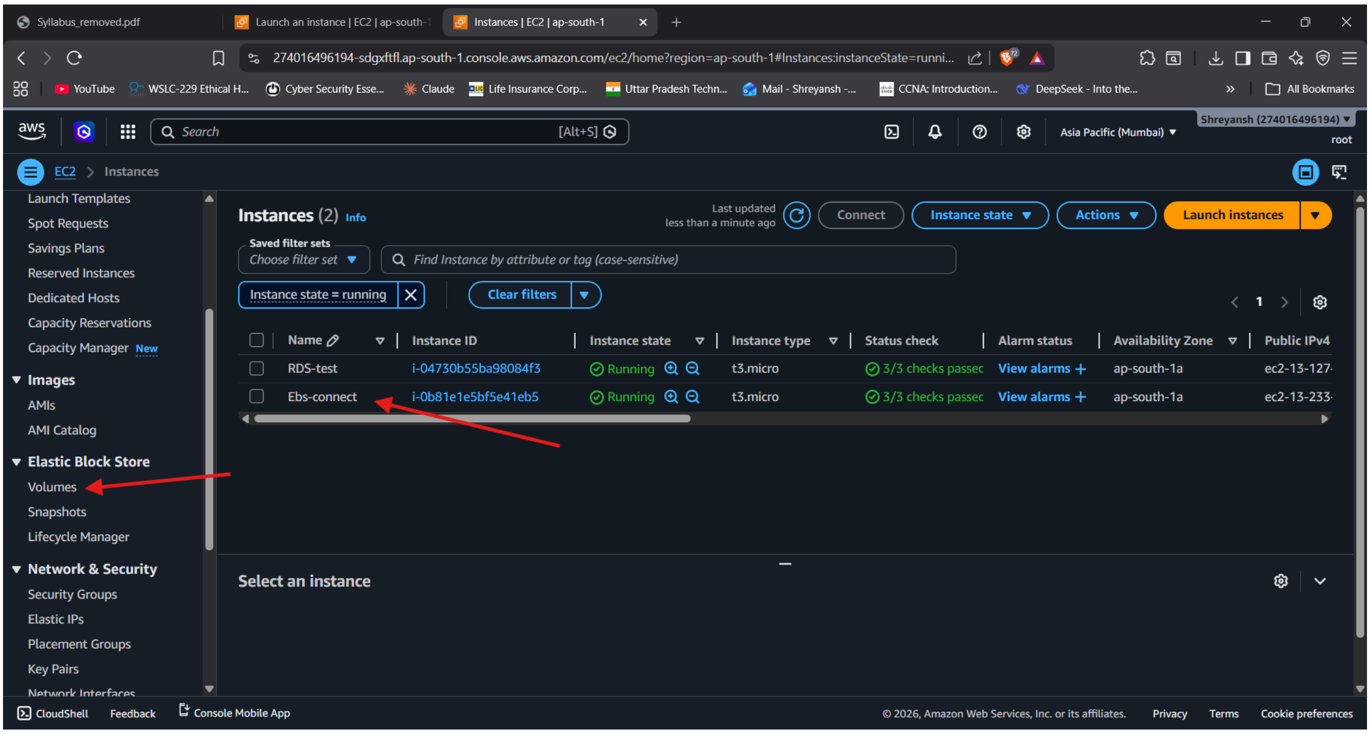
AWS confirmed that the instance i-0b81e1e5bf5e41eb5 launched successfully.



Screenshot 2026-05-15 234646.png

Step 7: Open Elastic Block Store Volumes

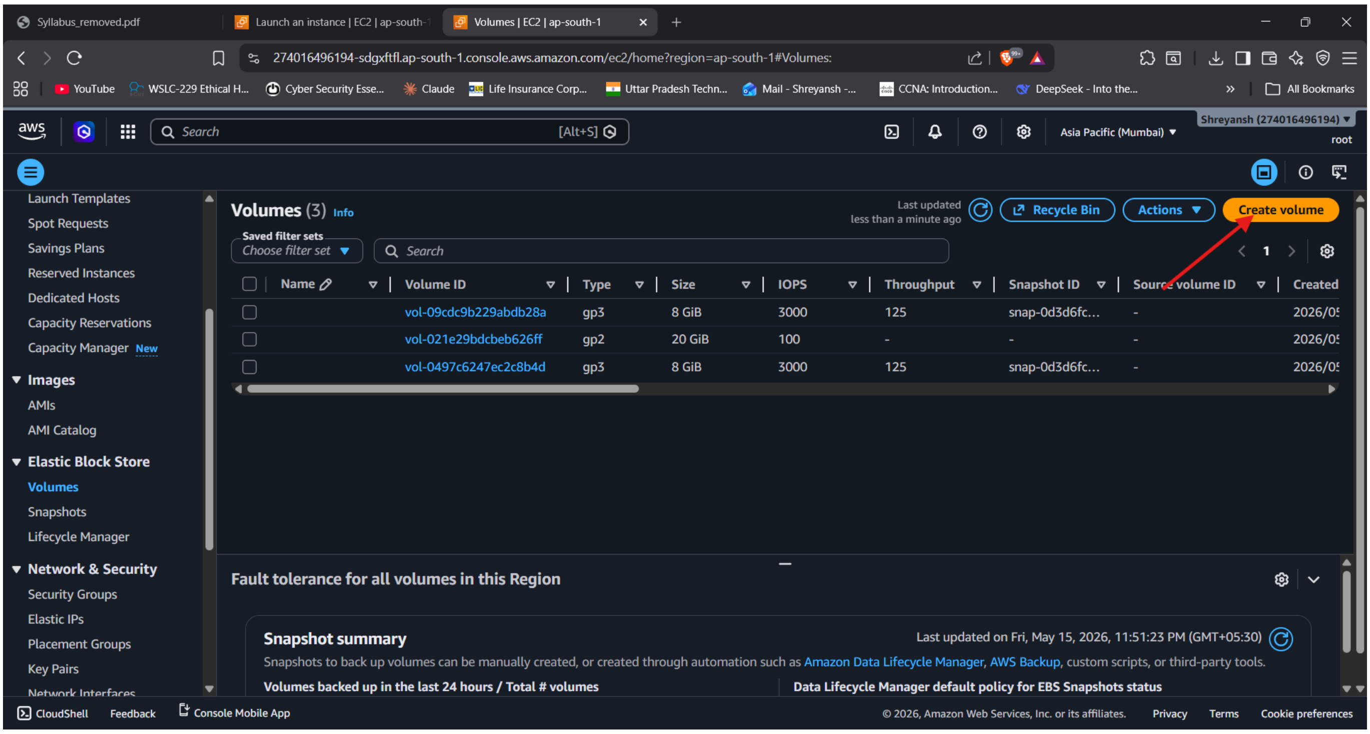
The Elastic Block Store Volumes section was opened to create an additional EBS volume.



Screenshot 2026-05-15 235024.png

Step 8: Create New EBS Volume

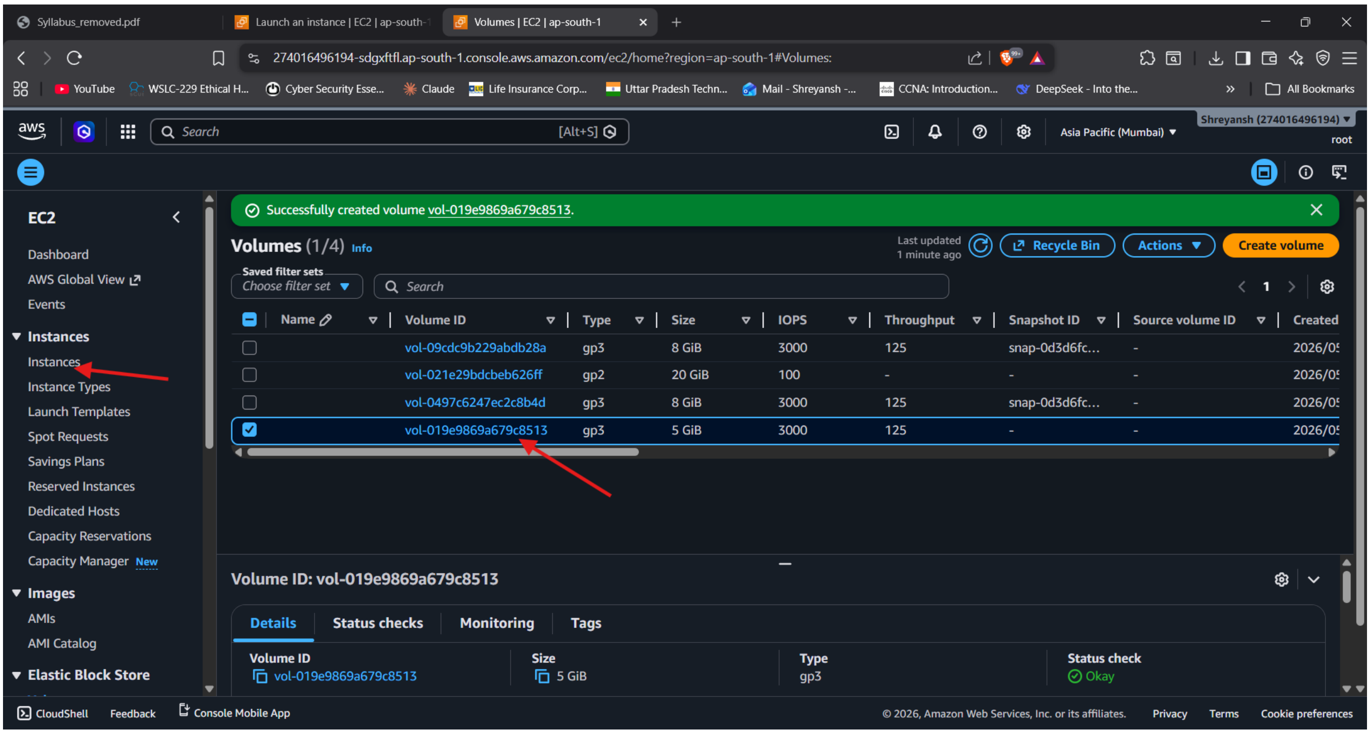
A General Purpose SSD (gp3) volume of 5 GiB was configured in availability zone ap-south-1a.



Screenshot 2026-05-15 235153.png

Step 9: Confirm Volume Creation

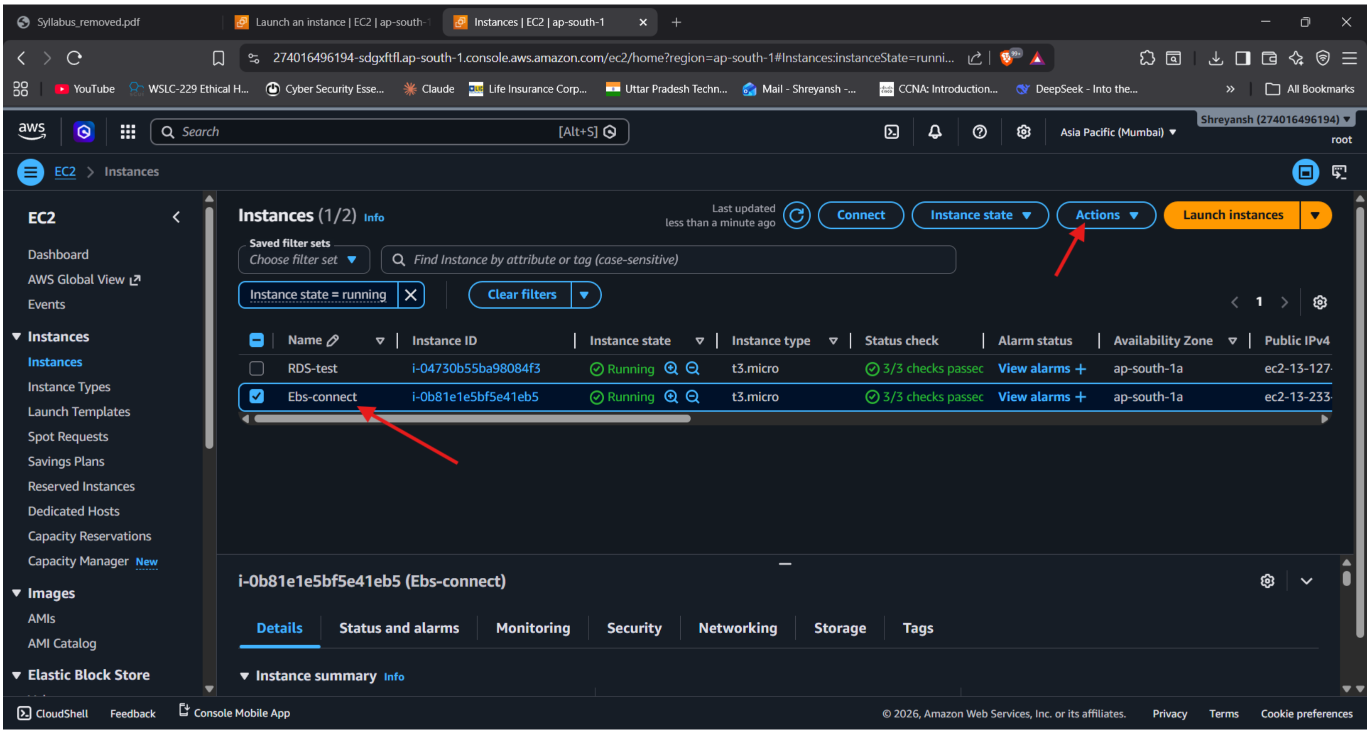
The new volume vol-019e9869a679c8513 was created successfully with 5 GiB size.



Screenshot 2026-05-15 235531.png

Step 10: Select EC2 Instance for Attachment

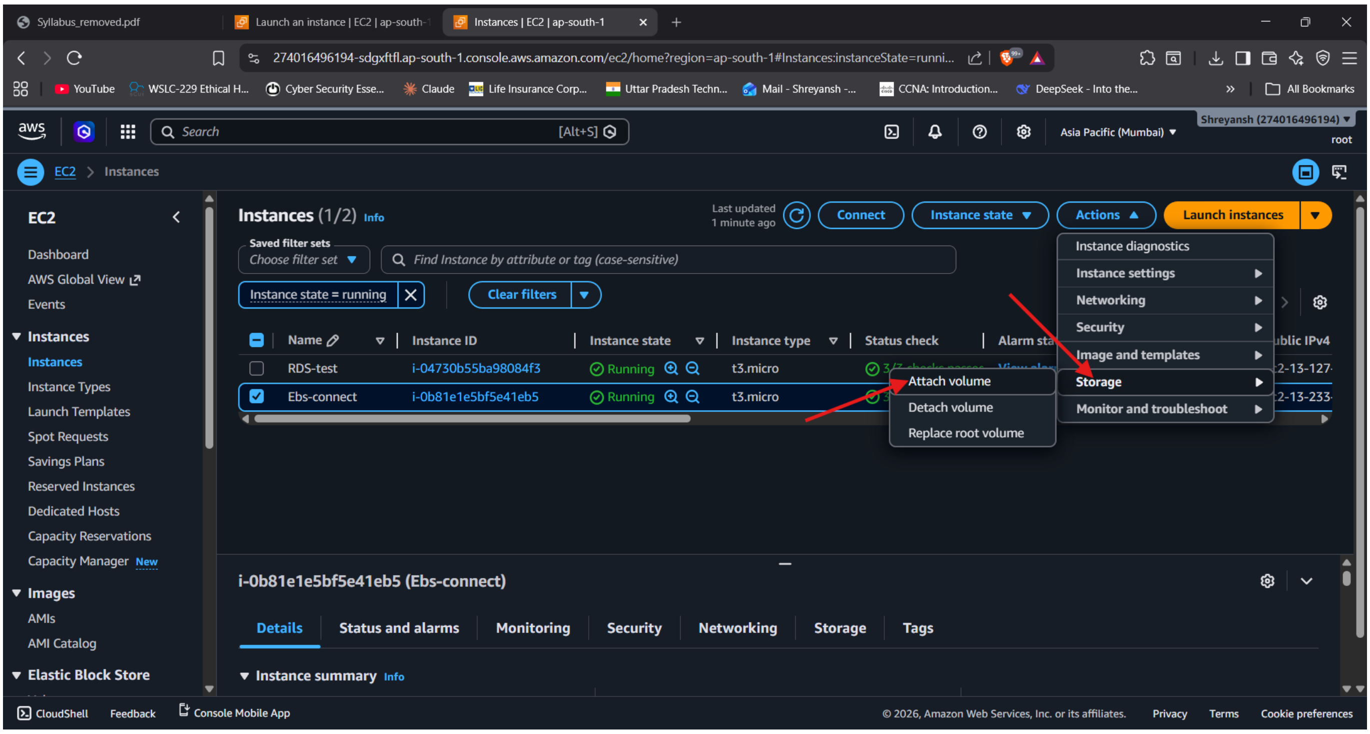
The Ebs-connect instance was selected, and the Storage > Attach volume action was used.



Screenshot 2026-05-15 235639.png

Step 11: Attach EBS Volume

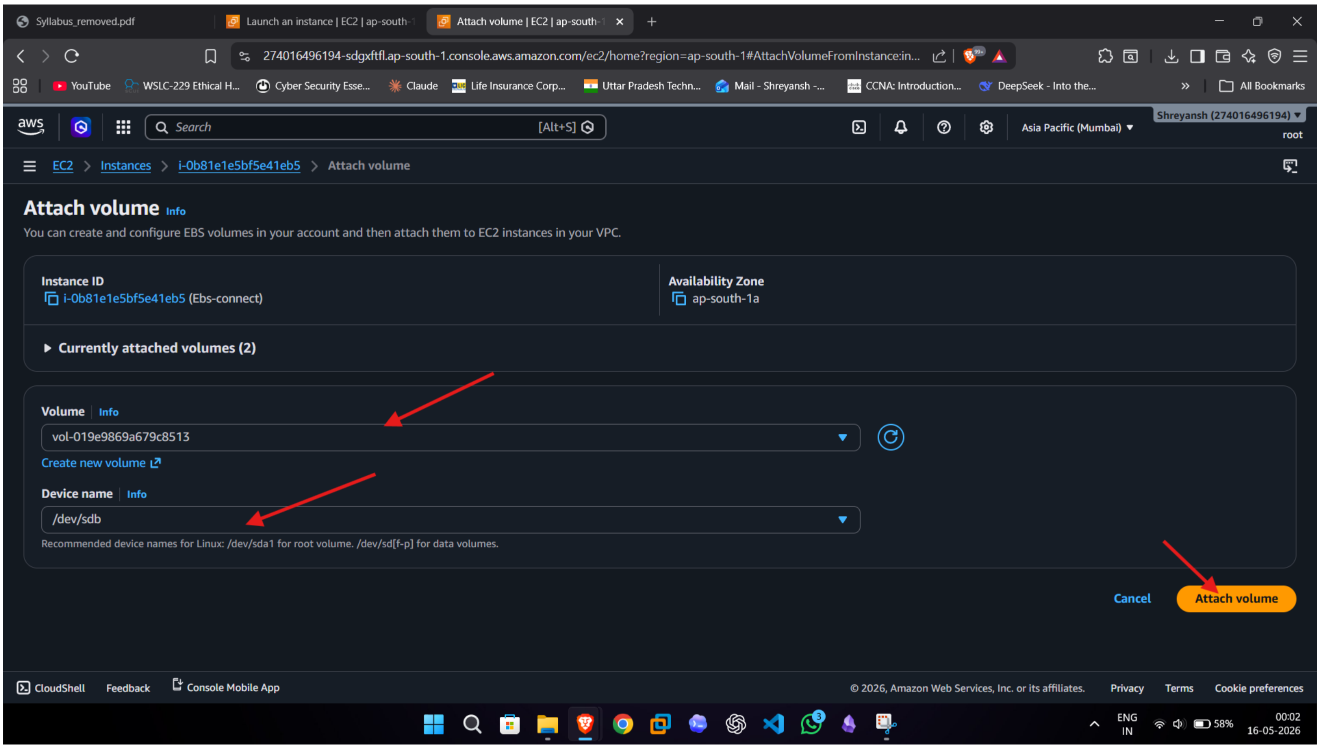
The EBS volume vol-019e9869a679c8513 was attached to the EC2 instance using a Linux device name.



Screenshot 2026-05-15 235715.png

Step 12: Verify Volume in AWS Storage Tab

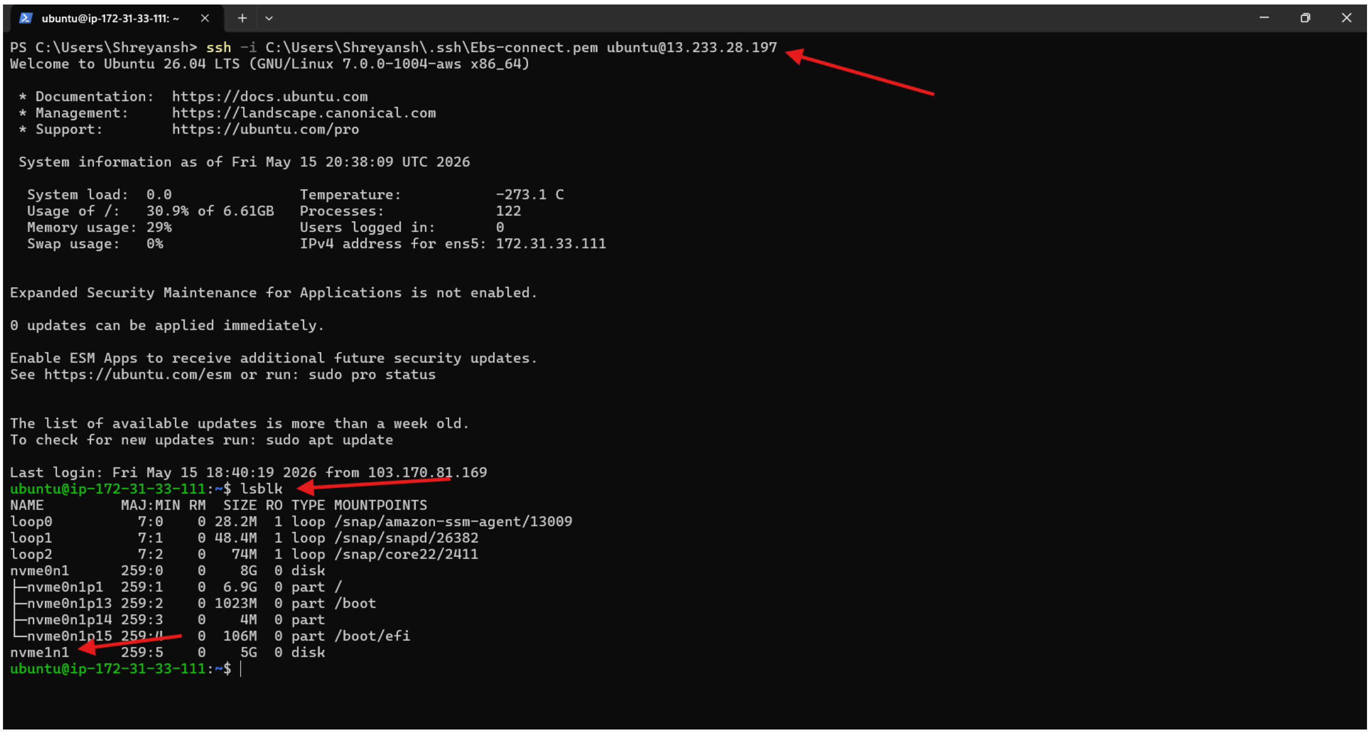
The Storage tab shows both the root volume and the newly attached 5 GiB EBS volume.



Screenshot 2026-05-16 000208.png

Step 13: Connect to EC2 and Check Disk

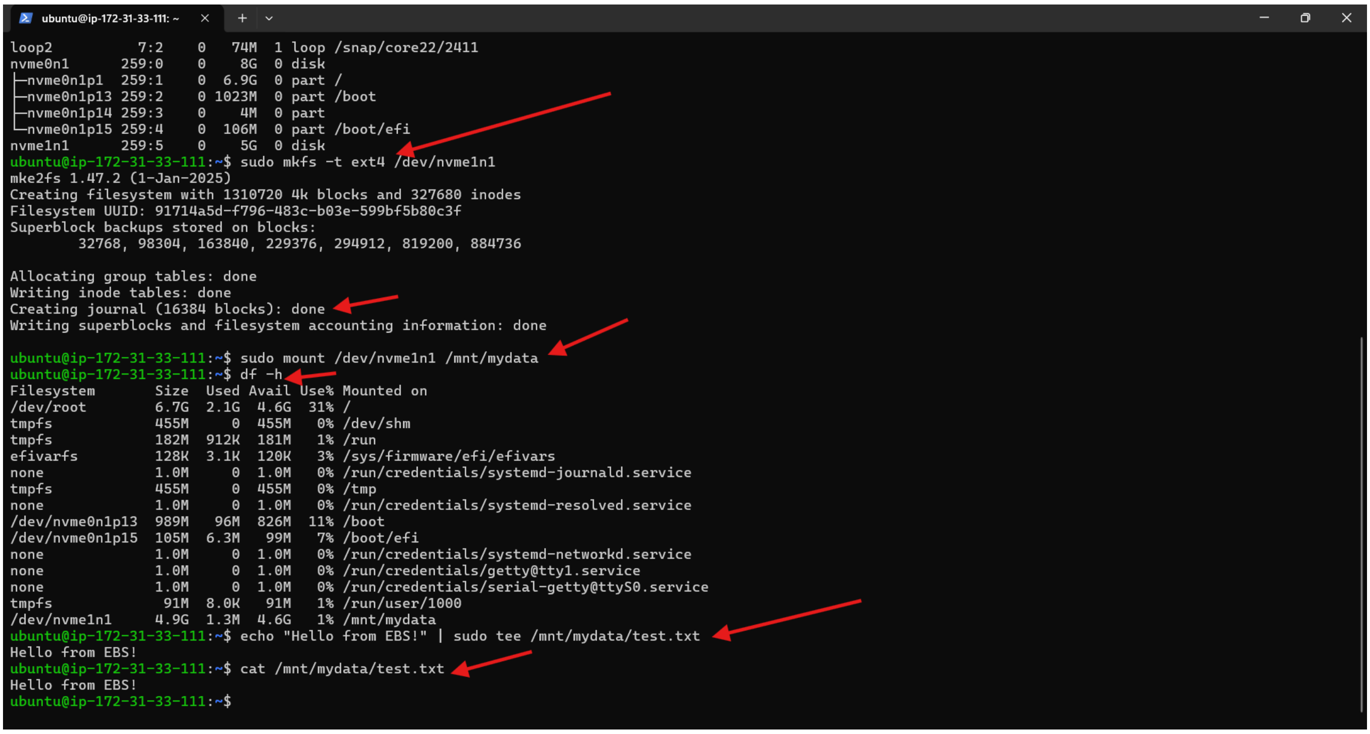
SSH connection was established from Windows Terminal and lsblk confirmed the new 5 GiB disk.



Screenshot 2026-05-16 020831.png

Step 14: Format, Mount, and Test EBS Volume

The EBS volume was formatted as ext4, mounted at /mnt/mydata, and tested by writing and reading a file.



Screenshot 2026-05-16 021416.png

Linux Commands Used

```
ssh -i C:\Users\Shreyansh\.ssh\Ebs-connect.pem ubuntu@13.233.28.197
lsblk
sudo mkfs -t ext4 /dev/nvme1n1
sudo mount /dev/nvme1n1 /mnt/mydata
df -h
echo "Hello from EBS!" | sudo tee /mnt/mydata/test.txt
cat /mnt/mydata/test.txt
```

Verification Result

The EC2 instance was launched successfully, the EBS volume was created and attached, and Ubuntu detected the 5 GiB disk as /dev/nvme1n1. After formatting the disk with ext4 and mounting it at /mnt/mydata, the test file was written and read successfully. This confirms that the EBS volume is attached, mounted, and usable for storage.

Check	Observed Result	Status
EC2 instance launch	Instance i-0b81e1e5bf5e41eb5 running	Successful
EBS volume creation	5 GiB gp3 volume vol-019e9869a679c8513 created	Successful
Volume attachment	Volume attached to Ebs-connect instance	Successful
Linux detection	lsblk shows /dev/nvme1n1 as 5G disk	Successful
Mount verification	df -h shows /mnt/mydata mounted	Successful
Read/write test	test.txt contains Hello from EBS!	Successful

Conclusion

This practical successfully demonstrates the complete AWS EC2 and EBS workflow: launching an Ubuntu EC2 instance, creating a separate EBS volume, attaching it to the instance, formatting and mounting it in Linux, and validating storage access with a read/write test.